Mephisto™ MOPA Laser

Operator's Manual

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Operator's Manual
Mephisto™ MOPA Laser

COHERENT.
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1 INTRODUCTION

1.1 Signal Words and Symbols in this Manual

This documentation may contain sections in which particular hazards are defined or special attention is drawn to particular conditions. These sections are indicated with signal words in accordance with ANSI Z-535.6 and safety symbols (pictorial hazard alerts) in accordance with ANSI Z-535.3 and ISO 7010.

1.1.1 Signal Words

Four signal words are used in this documentation: **DANGER**, **WARNING**, **CAUTION** and ***NOTICE***.

The signal words **DANGER**, **WARNING** and **CAUTION** designate the degree or level of hazard when there is the risk of injury:

DANGER!
Indicates a hazardous situation that, if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations.

WARNING!
Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

CAUTION!
Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.

The signal word "***NOTICE***" is used when there is the risk of property damage:

NOTICE
Indicates information considered important, but not hazard- related.

Messages relating to hazards that could result in both personal injury and property damage are considered safety messages and not property damage messages.

1.1.2 Symbols

The signal words **DANGER**, **WARNING**, and **CAUTION** are always emphasized with a safety symbol that indicates a special hazard, regardless of the hazard level:

This symbol is intended to alert the operator to the presence of additional information.

This symbol is intended to alert the operator to the presence of important operating and maintenance instructions.

This symbol is intended to alert the operator to the danger of exposure to hazardous visible and invisible laser radiation.

This symbol is intended to alert the operator to the presence of dangerous voltages within the product enclosure that may be of sufficient magnitude to constitute a risk of electric shock.

This symbol is intended to alert the operator to the danger of Electro-Static Discharge (ESD) susceptibility.

This symbol is intended to alert the operator to the danger of crushing injury.

This symbol is intended to alert the operator to the danger of a lifting hazard.

1.2 Preface

This manual contains user information for the Mephisto MOPA.

NOTICE
Read this manual carefully before operating the laser for the first time. Failure to follow the instructions and safety precautions in this manual can result in serious injury or death. Special attention must be given to the material in “Laser Safety” (p. 9), that describes the safety features built into the laser. Keep this manual with the product and in a safe location for future reference.

DANGER!
Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

1.3 Export Control Laws Compliance

It is the policy of Coherent to comply strictly with U.S. export control laws.

Export and re-export of lasers manufactured by Coherent are subject to U.S. Export Administration Regulations, which are administered by the Commerce Department. In addition, shipments of certain components are regulated by the State Department under the International Traffic in Arms Regulations.

The applicable restrictions vary depending on the specific product involved and its destination. In some cases, U.S. law requires that U.S. Government approval be obtained prior to resale, export or re-export of certain articles. When there is uncertainty about the obligations imposed by U.S. law, clarification must be obtained from Coherent or an appropriate U.S. Government agency.

Products manufactured in the European Union, Singapore, Malaysia, Thailand: These commodities, technology, or software are subject to local export regulations and local laws. Diversion contrary to local law is prohibited. The use, sale, re-export, or re-transfer directly or indirectly in any prohibited activities are strictly prohibited.

1.4 The Operator’s Manual

This Operator Manual is designed to familiarize the user with the Mephisto MOPA system and its designated use. It contains important information on how to install, operate, and troubleshoot the laser system safely, properly, and most efficiently. Observing these instructions helps to avoid danger, reduce repair costs, and downtimes and increase the reliability and lifetime of the laser system.

This Manual:

- describes the physical hazards related to the laser system, the means of protection against these hazards, and the safety features incorporated in the design of the laser system
- briefly describes the purpose and operation as well as the primary features, system elements, subsystems, and fundamental laser control routines of the laser system
- describes the fundamental operation of the laser system
- describes the maintenance procedures for the laser system which can be performed by the end user. This includes a time schedule for all periodic routine replacement procedures and a basic troubleshooting section.

The screenshots in this manual are only examples and may show configurations or parameter settings which do not apply to Mephisto system. Changing parameter settings to correspond with screenshots may reduce laser performance or even damage the laser system!

1.4.1 Intended Audience

The Operator's Manual is intended for all persons that are to work on or with the laser system. It assumes that the reader has received guidance from their company's laser safety officer on the safe operation of the laser system.

None of the procedures described in this manual requires the defeating of safety interlocks. Where specific training is required to perform procedures, this is clearly indicated at the beginning of the corresponding section.

1.4.2 Availability and Use

This Operator's Manual must always be available wherever the laser system is in use. Keep this manual in a safe location for future reference. It must be read and applied by any person in charge of carrying out work with and on the laser system, such as

- operation (including setting up, troubleshooting in the course of work, removal of production waste, care and disposal of consumables,
- service (maintenance, inspection, repair) and/or
- transport.

The sections are numbered continuously.sectionsEach step within a procedure is sequentially numbered. Each procedure starts with the step number one.

1.4.3 Cited Standards

Unless otherwise stated, all technical standards cited in this manual relate to the latest version of the standard that is applicable at the date of the publication of this manual.

This information is in compliance with the Performance Standards for Laser Products,' *United States Code of Federal Regulations*, 21 CFR 1040.10(d). In many cases, the international standards (ISO and IEC standards) have been adopted wholly or in part by national or regional standards authorities and are known locally under the designation assigned by this authority. For instance, the IEC 60825-1 has been adopted by the European Committee for Standardization as the standard EN 60825-1 and, in turn, by various national standards authorities as standards such as DIN EN 60825 (Germany) and BS EN 60825 (United Kingdom). The exact content, number and revision date of the national standard may, however, vary from that of the corresponding international standard. For further information, please contact the publisher of the respective national standard.

1.5 Laser Terminology

ISO 11145 (“Optics and Optical Instruments - Lasers and Laser Related Equipment - Vocabulary and Symbols”) contains a list of laser terminology.

To prevent misunderstandings, the Mephisto documentation strictly differentiates between “laser” and “laser system”?. Thus “start laser system” means that the power is off and shall be turned on. To “start the laser” means to switch on the laser beam and start laser operation.

A Laser
Consists of an amplifying medium capable of emitting coherent radiation with wavelengths up to 1 mm by means of stimulated emission.

B Laser system
A laser (A), where the radiation is generated, together with essential additional facilities (E) that are necessary to operate the laser (e.g. cooling, power, and gas supply).

In addition to the terminology used by ISO 11145, IEC 60825-1 uses the term “laser product”. This term relates to any product or assembly of components which constitutes or is intended to incorporate a laser. In other words, the term “laser product” can be used in conjunction with any of the definitions contained in ISO 11145.

1.6 Units of Measurements

In this manual, units of measurement are used according to the metric system (international system of units (SI)), e.g. meter, millimeter, square meter, cubic meter, liter, kilogram, bar, pascal; and imperial system, e.g. tons, pounds, and ounces; gallons and quarts; miles, yards, feet, and inch.

Temperatures are primarily indicated in degrees Celsius (°C) and Fahrenheit (°F).

1.7 Feedback Regarding Documentation

If there are any comments regarding the documentation provided, please contact the Coherent Documentation Department.

In any correspondence, please provide the following:

- the document part number, revision, and date of issue,
- the section number, page number and, where applicable, the procedure step number,
- a description of any errors,
- a proposal for improvements.

1.7.1 Feedback Address

E-mail documentation.support@coherent.com

Post Coherent Inc
Documentation and Training Development Department
5100 Patrick Henry Drive
Santa Clara, CA. 95054
USA

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2 LASER SAFETY

This user information is in compliance with the following standards for Light-Emitting Products IEC 60825-1 / EN 60825-1 *“Safety of laser products - Part 1: Equipment classification and requirements”* 21 CFR Title 21 Chapter 1, Subchapter J, Part 1040 *“Performance standards for light-emitting products”*.

WARNING!
LASER RADIATION - AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION CLASS 4 LASER PRODUCT!

WARNING!
Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

This laser safety section must be reviewed thoroughly prior to operating the Mephisto MOPA laser system. Safety instructions presented throughout this manual must be followed carefully.

2.1 Hazards

Hazards associated with lasers generally fall into the following categories:

- Biological hazards from exposure to laser radiation that may damage the eyes or skin
- Electrical hazards generated in the laser power supply or associated circuits
- Chemical hazards resulting from contact of the laser beam with volatile or flammable substances, or released as a result of laser material processing

The above list is not intended to be exhaustive. Anyone operating the laser must consider the interaction of the laser system with its specific working environment to identify potential hazards.

2.2 Optical Safety

Laser light, because of its optical qualities, poses safety hazards not associated with light from conventional light sources. The safe use of lasers requires all operators, and everyone near the laser system, to be aware of the dangers involved. Users must be familiar with the instrument and the properties of coherent, intense beams of light.

The safety precautions listed below are to be read and observed by anyone working with or near the laser. At all times, ensure that all personnel who operate, maintain or service the laser are protected from accidental or unnecessary exposure to laser radiation exceeding the accessible emission limits defined in the laser safety standards.

WARNING!
Direct eye contact with the output beam from the laser may cause serious damage and possible blindness.

The greatest concern when using a laser is eye safety. In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces such as lenses or beamsplitters. While weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing, or combustible materials, even at some distance. They can ignite volatile substances such as alcohol, gasoline, ether, and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers, and photodiodes. The user is advised to follow the control measures below.

2.2.0.1 Recommended Precautions and Guidelines

1. Observe all safety precautions in the operator's manual.
2. Always wear appropriate eyewear for protection against the specific wavelengths and laser energy being generated. See “Laser Safety Eyewear” (p. 12) for additional information.

3. Avoid wearing watches, jewelry, or other objects that may reflect or scatter the laser beam.
4. Stay aware of the laser beam path, particularly when external optics are used to steer the beam.
5. Provide enclosures for beam paths whenever possible.
6. Use appropriate energy-absorbing targets for beam blocking.
7. Block the beam before applying tools such as Allen wrenches or ball drivers to external optics.
8. Limit access to the laser to trained qualified users who are familiar with laser safety practices. When not in use, lasers should be shut down completely and made off-limits to unauthorized personnel.
9. Terminate the laser beam with a light-absorbing material. Laser light can remain collimated over long distances and therefore presents a potential hazard if not confined. It is good practice to operate the laser in an enclosed room.
10. Post laser warning signs in the area of the laser beam to alert those present.
11. Exercise extreme caution when using solvents in the area of the laser.
12. Never look directly into the laser light source or at scattered laser light from any reflective surface, even when wearing laser safety eyewear. Never sight down the beam.
13. Set up the laser so that the beam height is either well below or well above eye level.
14. Avoid direct exposure to the laser light. Laser beams can easily cause flesh burns or ignite clothing.
15. Advise all those working with or near the laser of these precautions.

CAUTION!
Laser safety eyewear protects the user from accidental exposure to laser radiation by blocking light at the laser wavelengths. However, laser safety eyewear may also prevent the operator from seeing the beam or the beam spot. Exercise extreme caution even while wearing safety glasses.

WARNING!
Do not modify the laser and do not open the laser housing! Doing so could lead to exposure to additional radiation at a wavelength of 0.8-0.82 µm and could also lead to damage to the laser.

2.2.1 Laser Safety Eyewear

Always wear appropriate laser safety eyewear for protection against the specific wavelengths and laser energy being generated. The appropriate eye protection can be calculated as defined in the “EN 207 Personal eye protection equipment - Filters and eye-protectors against laser radiation (laser eye-protectors)”, in other national or international standards (e.g. ANSI, ACGIH, or OSHA) or as defined in national safety requirements.

CAUTION!
Laser safety eyewear protects the user from accidental exposure to laser radiation by blocking light at the laser wavelengths. However, laser safety eyewear may also prevent the operator from seeing the beam or the beam spot. Exercise extreme caution even while wearing safety glasses.

2.2.1.1 Viewing Distance (Nominal Ocular Hazard Distance)

The Mephisto MOPA laser produces optical power levels that are dangerous to the eyes and skin if exposed directly or indirectly. This product must be operated only with proper eye and skin protection at all times. Never view directly emitted or scattered radiation with unprotected eyes. When viewing the laser during operation, the operator must maintain the Nominal Ocular Hazard Distance (NOHD) between the laser or scatter radiation and the operator's eyes. The maximum NOHD calculation (maximum power plus safety margin as specified on the danger label) for the Mephisto MOPA is 1200 m as specified in IEC 60825-1.

2.3 Electrical Safety

Mephisto MOPA lasers use AC and DC voltages. There are no user serviceable components in the controller or laser head. All units are designed to be operated as assembled. Warranty will be voided if the laser head, the controller, or the cable is disassembled.

DANGER!
Normal operation of the Mephisto MOPA laser should not require access to the power supply circuitry. Removing the power supply cover will expose the user to potentially lethal electrical hazards. Do not open the power supply. Only authorized service personal at Coherent should attempt to correct any problem with the power supply.

2.3.0.1 Recommended Precautions and Guidelines

The following precautions must be observed by everyone when working with potentially hazardous electrical circuitry:

DANGER!
When working with electrical power systems, the rules for electrical safety must be strictly followed. Failure to do so could result in the exposure to lethal levels of electricity.

1. Disconnect main power lines before working on any electrical equipment when it is not necessary for the equipment to be operating.
2. Do not short or ground the power supply output. Protection against possible hazards requires proper connection of the ground terminal on the power cable, and an adequate external ground. Check these connections at the time of installation, and periodically thereafter.
3. Never work on electrical equipment unless there is another person nearby who is familiar with the operation and hazards of the equipment, and who is competent to administer first aid.
4. When possible, keep one hand away from the equipment to reduce the danger of current flowing through the body if a live circuit is touched accidentally.
5. Always use approved, insulated tools.
6. Special measurement techniques are required for this system. A technician who has a complete understanding of the system operation and associated electronics must select ground references.
7. Do not use liquids, including water, on the laser head or on the electronics.

2.4 Maximum Accessible Radiation Level and Intended Use

The Mephisto MOPA laser produces invisible radiation over a wavelength range of 1.06 to 1.07 µm, with a maximum of 70 W continuous wave power.

The radiation provided by this laser system must be used in an appropriate environment and all necessary safety precautions must be observed.

2.5 Safety Features and Compliance with Government Requirements

The following features are incorporated into the instrument to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, Subchapter J, Part 1040 administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the Low Voltage Directive (LVD) (published in 2006/95/EC). The Low Voltage Directive requires that lasers comply with the standard EN 61010-1/IEC 61010-1 "Safety Requirements for Electrical Equipment for Measurement, Control and Laboratory Use" and EN 60825-1/IEC 60825-1 "Safety of Laser Products". Compliance of this laser with the LVD requirements is certified by the CE mark.

2.5.1 Laser Classification

Governmental standards and requirements specify that the laser must be classified according to the output power or energy and the laser wavelength. The Mephisto MOPA laser is classified as Class 4 based on 21 CFR, Subchapter J, Part 1040, section 1040.10 (c) and/or IEC/EN 60825-1, Clause 5. In this manual, the classification will be referred to as Class 4.

2.5.2 Protective Housing

The laser head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class 1 radiation as specified in the 21 CFR, Part 1040 Section 1040.10 (f)(1) and EN 60825-1/IEC 60825-1 Clause 6.2 except for the output beam, which is Class 4.

2.5.3 Remote Laser Disable Connector

Both the amplifier and the master laser controller include a remote disable that functions as an external interlock connector. This connector allows the laser to be disabled remotely. This is not a redundant safety interlock.

The left laser disable pin of the master laser controller must be connected to the left laser disable connector of the amplifier controller and the two center pins should also be connected to each other. This way both controllers can be shut down at the same time, thereby shutting down the master and the amplifier. A passive potential free switch must be connected to these pins. The water cooling interlock should be connected to this laser disable to shut down the Mephisto MOPA if necessary. See Figure 2-1.

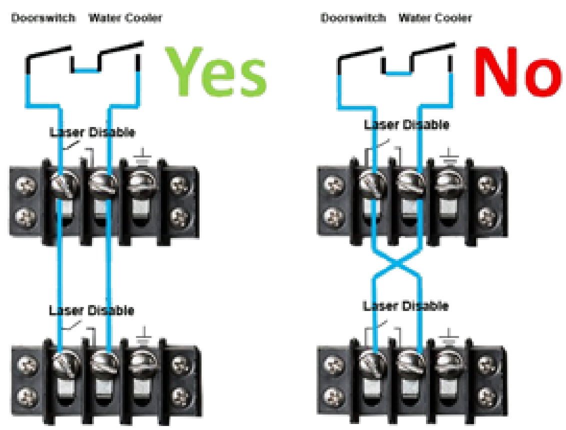


Figure 2-1. Connecting the Laser Disable

2.5.4 Key Control (Power Supply)

Operation of the Mephisto MOPA laser requires that the power supply keyswitch be in the ON position. The key is removable and the system cannot be operated when the key is removed. The key can not be removed in the ON position, so that the system can always be turned off. [CFR 1040.10 (f)(4)/EN 60825-1/IEC 60825-1, Clause 6.6].

2.5.5 Laser Emission and Classification

The LASER EMISSION indicators on both the power supply and the laser indicate laser emission. Make sure that they are visible through the laser safety glasses.

2.5.6 Beam Attenuator

An internal shutter prevents exposure to all laser radiation without removing power from the system [CFR 1040.10 (f)(6)/EN 60825-1/ IEC 60825-1, Clause 6.8].

2.5.7 Operating Controls

The laser controls are positioned so that the operator is not exposed to laser emission while manipulating the controls [CFR 1040.10(f)(7)/EN 60825-1/IEC 60825-1, Clause 6.9].

2.5.8 Display Screen

The display screen on the front of the panel of the power supply may be viewed without exposing the operator to the laser emission [CFR 1040.10(f)(8)/EN 60825-1/IEC 60825-1, Clause 4.10].

WARNING!
Use of controls or adjustments or performance of procedures other than those specified in the manual may result in hazardous radiation exposure.

NOTICE
Use of the system in a manner other than that described herein may impair the protection provided by the system.

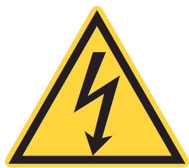
2.6 Electromagnetic Compatibility

The European requirements for Electromagnetic Compliance (EMC) are specified in the EMC Directive (published in 2014/30/EU).

Class A: Conformance (EMC) concerning emission and immunity is achieved through compliance with the harmonized standard EN 61326-1:2013 (Electrical Requirement for Measurement, Control and Laboratory) for Class A.

The laser meets the emission requirements for Class A, group 1 as specified in EN55011.

Compliance of this laser with the EMC requirements is certified by the CE mark.



WARNING!
To avoid damage to the laser or bodily injury, check that the line voltage does not exceed the limits of the electronic. Also check that the environmental conditions and the use of the modulation inputs are as described herein.

WARNING!
This laser can cause serious bodily injury and must not be used for any medical applications whatsoever.

2.7 Environmental Compliance

2.7.1 RoHS Compliance

The RoHS directive restricts the use of certain hazardous substances in electrical and electronic equipment. Coherent can provide RoHS certification upon request for products requiring adherence to the RoHS Directive.

All components of the Mephisto MOPA laser system are RoHS compliant.

Compliance of this laser with the EMC requirements is certified by the CE mark.

2.7.2 China-RoHS Compliance

The China-RoHS directive restricts the use of certain hazardous substances in electrical and electronic equipment and applies to the production, sale, and import of products in the Peoples Republic of China. Refer to the Table 2-1 for product components that are China-RoHS compliant.

Table 2-1. China-RoHS Compliant Components

部件名称 Part Name	产品中有害物质的名称及含量 有害物质 Hazardous Substances 铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr(VI))	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
印刷电路板组装 Printed Circuit Board Assembly	X	O	O	O	O	O
电缆装配 Cable Assembly	X	O	X	O	O	O
光学部件装配 Optic Assembly	X	O	X	O	O	O
板金组装 Sheet Metal Assembly	O	O	O	O	O	O
电源 Power Supply	O	O	O	O	O	O
组装二极管激光器 Laser Diode Assembly	O	O	O	O	O	O

本表格依据 SJ/T 11364 的规定编制

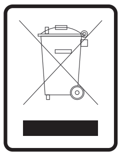
O: 表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。

X: 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 规定的限量要求。

2.7.3 Waste Electrical and Electronic Equipment (WEEE)

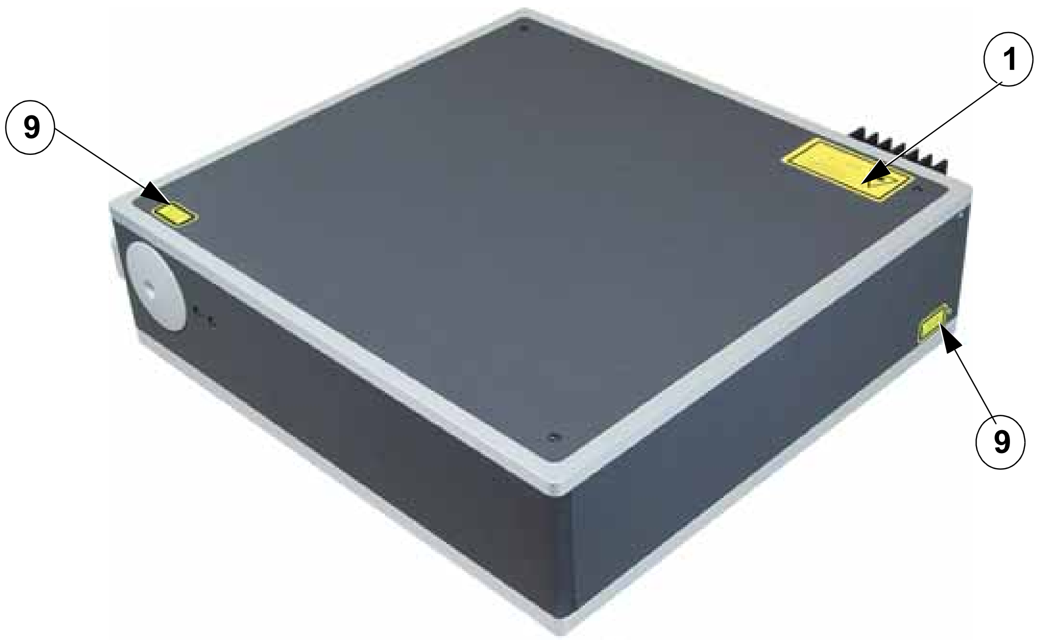
The European Waste Electrical and Electronic Equipment (WEEE) Directive (2012/19/EU) is represented by a crossed-out garbage container label. The purpose of this directive is to minimize the disposal of WEEE as unsorted municipal waste and to facilitate its separate collection. This icon is incorporated into the laser head and separate controller labels. See Figure 2-2 and Table 2-2.

The WEEE Directive applies to this product and any peripherals marked with this symbol. Do not dispose of these products as unsorted municipal waste. Contract the local distributor for procedures for recycling this equipment.



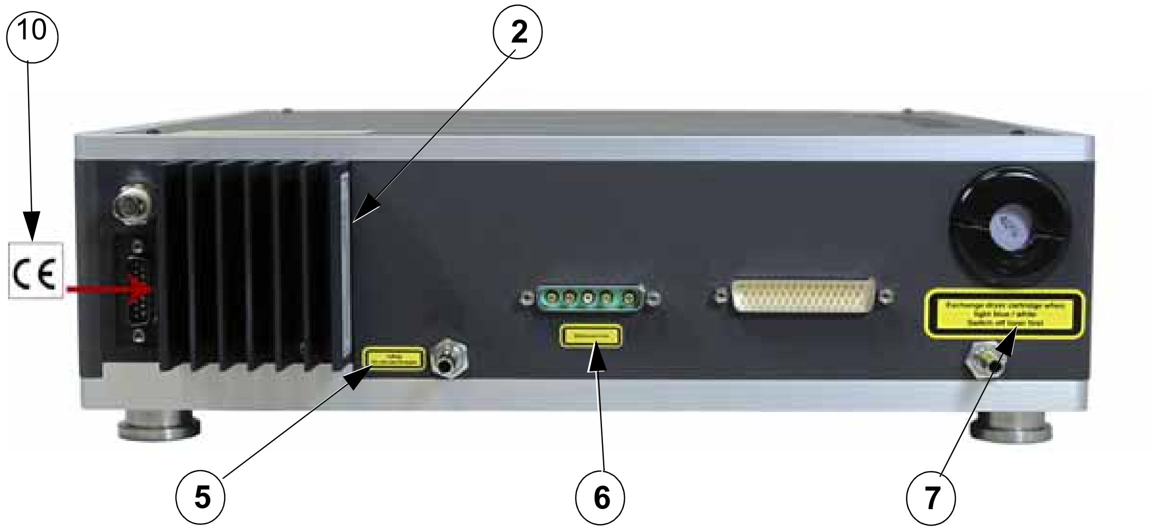
2.8 System Labels

Figure 2-2 identifies the location and Table 2-2 describes the labels found on the laser system. These include warning labels indicating removable or displaceable protective housings, apertures through which laser radiation is emitted, and labels of certification and identification [21 CFR § 1040.10(g), 21 CFR § 1010.2, and 21 CFR § 1010.3/ EN 60825-1/IEC 60825-1, Clause 7].



MOPA Laser Head - Top and Side View

Figure 2-2. Label Locations



MOPA Laser Head - Rear View

Figure 2-2. Label Locations (Continued)

Table 2-2. Label Descriptions

Item	Label	Description
1		**Radiation Danger Label for the Mephisto MOPA:** This label describes the specific wavelength and output power level capabilities of the laser system. It also includes the laser class of the product.
2	 Example	**Identification Label for the Laser Head:** This label contains the manufacturing date, model, serial number, and product origin. Also contains the WEEE symbol that is described in "Waste Electrical and Electronic Equipment (WEEE)" (p. 18).
3	 Example	**Identification Label for the Master Laser Controller:** This label contains the manufacturing date, model, serial number, and product origin. Also contains the WEEE symbol that is described in "Waste Electrical and Electronic Equipment (WEEE)" (p. 18).

Table 2-2. Label Descriptions (Continued)

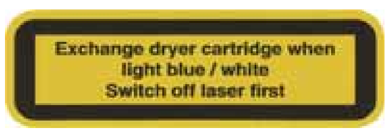
Item	Label	Description
4	 Example	**Identification Label for the Amplifier Laser Controller:** This label contains the manufacturing date, model, serial number, and product origin. Also contains the WEEE symbol that is described in "Waste Electrical and Electronic Equipment (WEEE, 2002)" on page 1-5.
5		Label for Water Inflow Connector
6		Label for Multiple Power Feeds
7		Label for Dryer Cartridge
8		**Laser Aperture Label:** This label identifies the location of the output beam.

Table 2-2. Label Descriptions (Continued)

Item	Label	Description
9		**Danger Label:** This label warns of laser radiation when the housing is opened.
10		**CE Label:** This label identifies CE certification and compliance.

2.9 Maintenance of Laser Safety Features

Verify the operation of all features listed below, either annually or whenever the product has been subjected to adverse environmental conditions (e.g. fire, flood, mechanical shock, spilled solvents, etc.).

1. Verify that all warning labels listed in Table 2-2 are present and firmly affixed in the correct locations.
2. Verify that opening the laser disable prevents laser operation.
3. Verify, while the mechanical shutter is closed, that the laser emission LED at the front of the laser head is operating.

2.10 Sources of Additional Information

The following are sources for additional information on laser safety standards and safety equipment and training.

2.10.1 Laser Safety Standards

Safe Use of Lasers
Document Z136.1
American National Standards Institute (ANSI)
www.ansi.org

Guidelines for Laser Safety and Hazard Assessment
Directives PUB 8-1.7
Occupational Safety and Health Administration (OSHA)
U.S. Department of Labor
www.osha.gov

A Guide for Control of Laser Hazards
American Conference of Governmental and Industrial Hygienists (ACGIH)
www.acgih.org

Laser Safety Guide
Laser Institute of America
www.lia.org

BSI Publications
British Standards Institution (BSI)
www.bsigroup.com

2.10.2 Equipment and Training

Laser Focus Buyer's Guide
Laser Focus World
www.laserfocusworld.com

Photonics Spectra Buyer's Guide
Photonics Spectra
www.photonics.com

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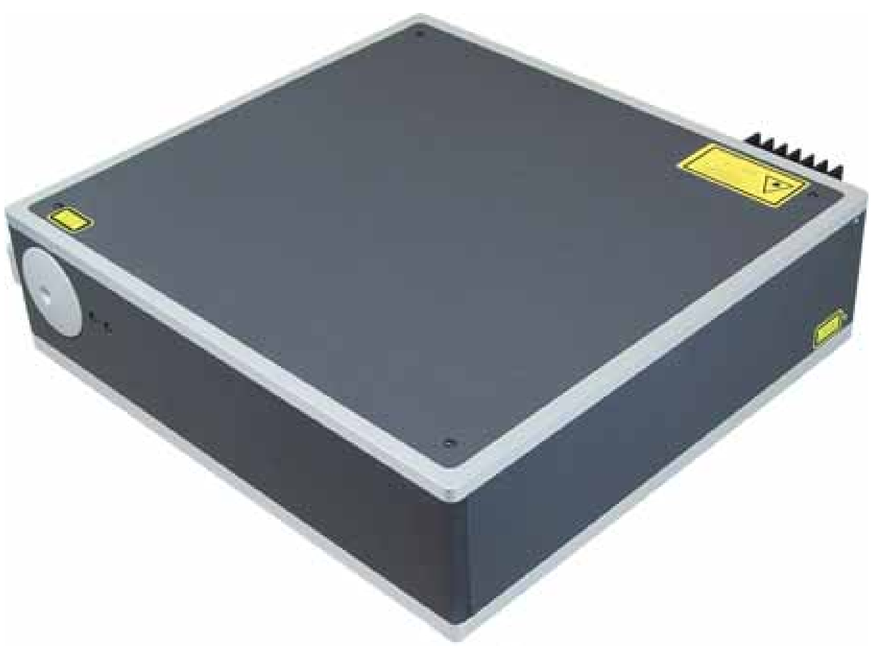
3 DESCRIPTION AND SPECIFICATIONS

3.1 Introduction

WARNING!
The Mephisto MOPA laser system is intended to be used in scientific and industrial measurement applications. DO NOT use this laser for any medical applications whatsoever as it can cause serious bodily injury.

The Mephisto MOPA laser system (Figure 3-1 on page 3-25) consists of three self-contained units: laser head, master laser controller, and amplifier controller . This manual is intended to provide detailed information on how to operate these devices properly.



Laser Head

Figure 3-1. Mephisto MOPA Laser System

Master Laser Controller

Amplifier Controller

Figure 3-1. Mephisto MOPA Laser System (Continued)

The laser head consists of a master laser and one or more amplifier stages. The master laser includes a temperature stabilized pump diode, providing the pump radiation for the monolithic Nd:YAG laser crystal. The monolithic design ensures excellent frequency stability of the emitted laser radiation. In addition, the laser crystal is temperature stabilized to increase the frequency stability even further. A fraction of the output power is directed onto a photo detector and the generated signal, appropriately filtered and amplified, is fed back to the pump diode. This feature is called Noise Eater. Activating this feedback loop, the Nd:YAG laser's intensity noise is suppressed by a large amount close to the fundamental limit set by quantum noise. The master laser is followed by one, two, three, or four amplifier stages to achieve the desired output power. Each amplifier stage has a temperature stabilized pump diode, pumping an amplifier.

In order to operate the Mephisto MOPA laser system, the laser head must be connected to an existing water-cooling system to the master laser controller and the amplifier controller. The amplifier output power depends on the temperature of the amplifier crystals, so that temperature fluctuation of the cooling water can be seen directly in the output power of the laser.

The controllers are designed to provide all required subsystems to drive and control the Mephisto MOPA laser system.

3.2 Specifications

The general specifications that apply to all Mephisto MOPA lasers are summarized in Table 3-1.

For customized laser parameters, please check the data sheet.

Table 3-1. General Specifications of the Mephisto MOPA Product Line

Parameter	Specification
Operational mode	Continuous wave
Wavelength/nm	1064
Polarization, extinction ratio	Linear horizontal, > 100:1
Beam quality	TEM00, M^2 < 1.3
Beam parameter product / mm·mrad[1]	< 0.44
Beam radius at the beam waist ($1/e^{2}$) / µm[1]	225
Beam waist location inside laser head (MOPA 8) / cm[1]	13.5
Beam waist location inside laser head (MOPA 25) / cm[1]	5.5
Beam waist location outside laser head (MOPA 42) / cm[1]	2.5
Beam waist location outside laser head (MOPA 55) / cm[1]	2.5
Divergence angle (half angle) / mrad[1]	1.7
Beam roundness	< 1.1
Thermal tuning coefficient laser crystal / GHz/K	- 3
Thermal tuning range laser crystal / GHz	30
Thermal response bandwidth laser crystal / Hz	≅ 1
Piezo tuning coefficient [MHz/V]	≅ 1
Piezo tuning range [MHz]	± 65
Piezo response bandwidth / kHz	100
Optical emission spectrum	single-frequency
Spectral linewidth / kHz / 100 ms	1
Coherence length / km	> 1
Frequency drift at constant room temperature / MHz/min	≅ 1
Relative Intensity Noise (RIN) > 20 kHz / dB/Hz	< -130
Laser head weight (MOPA 8,25,42) / kg	26
Laser head weight (MOPA 55) / kg	28
Master laser controller weight / kg	4.1
Master laser controller size / mm	340, 235, 90
Amplifier controller weight / kg	12.8

Table 3-1. General Specifications of the Mephisto MOPA Product Line (Continued)

Parameter	Specification
Amplifier controller size / mm	420, 450, 135
Laser head cooling requirements	water-cooling
Recommended water pressure / bar	4
Power variation with water temperature / %/K	< - 0.5
Supply current	single phase 50/60 Hz
Supply voltage master laser controller	110 to 240 V
Supply voltage amplifier	115/230 V
Fuse for 115 V	6.3 x 32 mm 16 A / 250 V slow blow
Fuse for 230 V	6.3 x 32 mm 8 A / 250 V slow blow
Supply current fluctuations / %	< |±10|
Controller housing IP Code	IP20
Laser cable minimum bending radius of curvature / mm	> 65
Typical electrical power consumption during operation (master laser) / W	40
Typical electrical power consumption during operation (amplifier MOPA 55W) / W	630
Maximum power consumption (master laser) / VA	150
Maximum power consumption (amplifier) / VA	1500
Environmental temperature range during operation / °C	+ 15 to + 30
Air pressure during operation / hPa	> 700
Clean room requirement	ISO 8
Humidity during operation	Non condensing
Vibration (sweeps 5 to 55 Hz @ 1 octave/min, for 3 axis) non operating / g	1
Environmental temperature range during transport/storage / °C	- 10 to + 50
Air pressure during transport/storage / hPa	> 200
Humidity during transport/storage	Non condensing
[1] These are average values. The lens positions inside the laser are optimized for maximum output power, therefore beam parameters can vary from one Mephisto laser to the next.	

3.3 Emission Spectrum

A common feature of all models of the Mephisto MOPA Product Line is the reliable emission on a single longitudinal mode and frequency. This can be investigated using an optical spectrum analyzer like a confocal scanning Fabry-Pérot Interferometer (FPI). The free spectral range (FSR) of the device was 2 GHz as indicated in Figure 3-2:

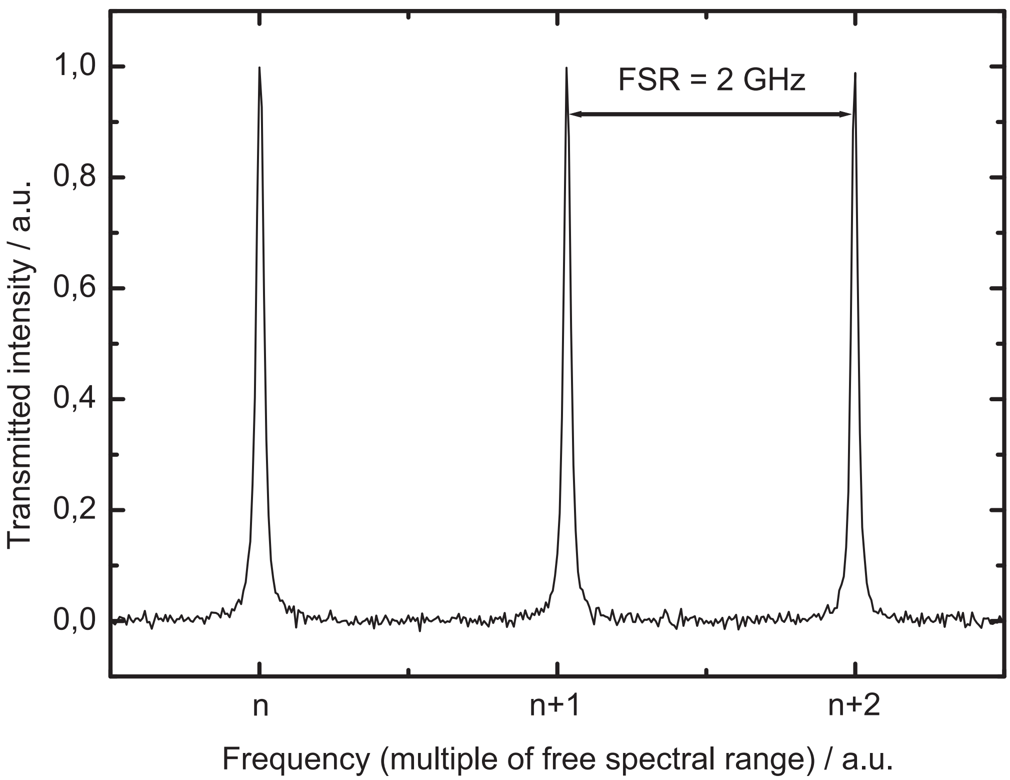


Figure 3-2. Emission Spectrum Using a Scanning Fabry-Pérot Interferometer (FPI)

The absence of any peaks between the main resonances of the interferometer clearly indicates the operation on a single longitudinal mode.

3.4 Relative Intensity Noise (RIN)

The Mephisto MOPA is equipped with an integrated intensity noise reduction system. This reduces intensity fluctuations of the laser beam by several orders of magnitude. These fluctuations are largely due to a phenomenon called relaxation oscillation.

The spectrum of these fluctuations resembles the spectral behavior of a white-noise-excited classical oscillator. It features a large peak and a significant amount of low frequency noise. This is illustrated in the trace "Free running" in Figure 3-3.

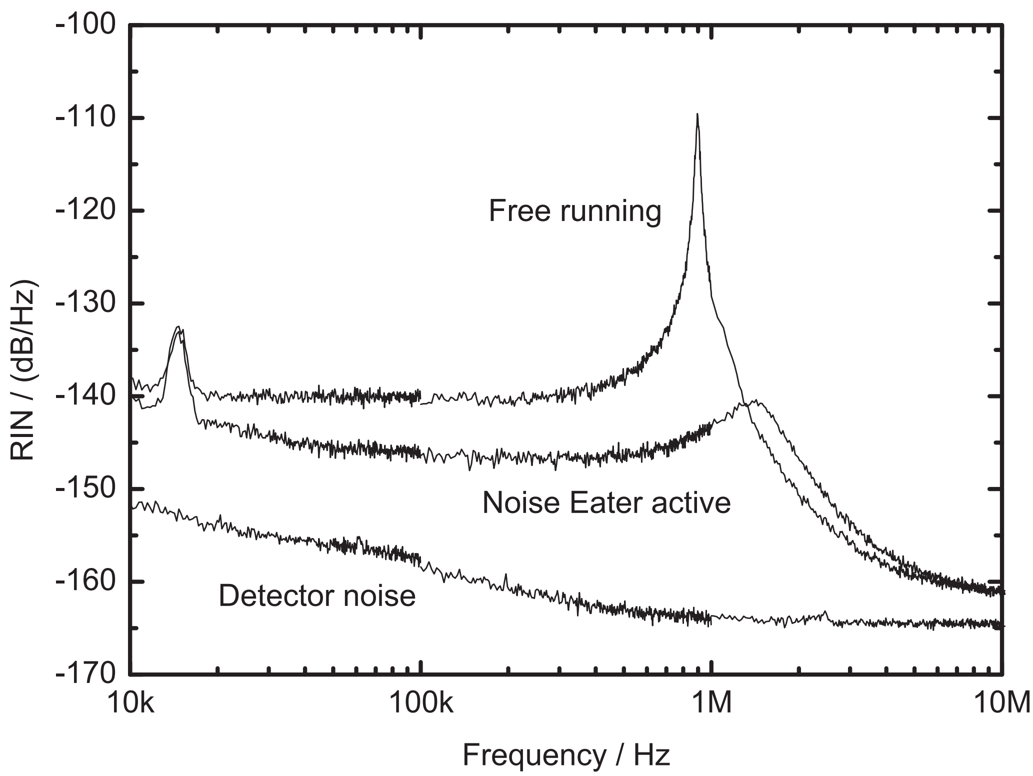


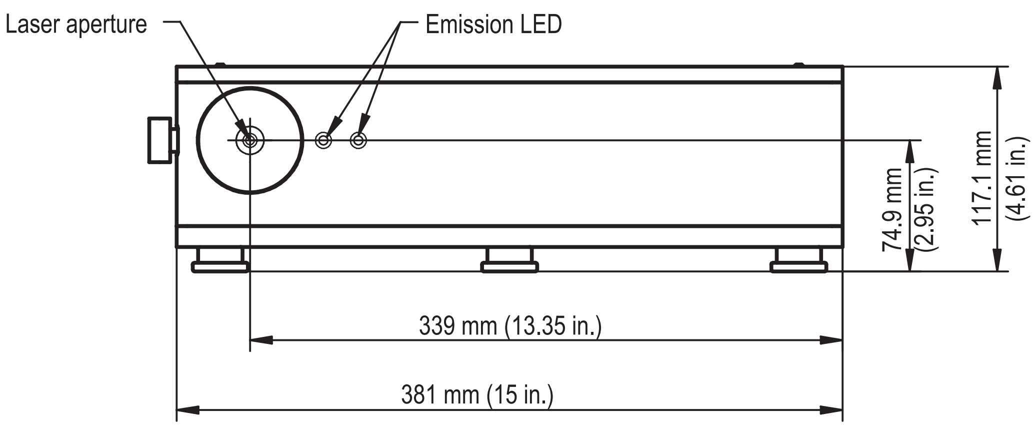
Figure 3-3. Relative Intensity Noise Measurements with and without Noise Eater Option

Both of these spectral components can be significantly reduced by using the Noise Eater option as illustrated by the trace "Noise Eater active" in Figure 3-3. To achieve this, a fraction of the generated Nd:YAG laser light, transmitted through a mirror, is directed onto a photo detector to analyze the variation of the generated output power. This signal, appropriately filtered and amplified by an electronic board, is fed to the laser diode current to stabilize the output power.

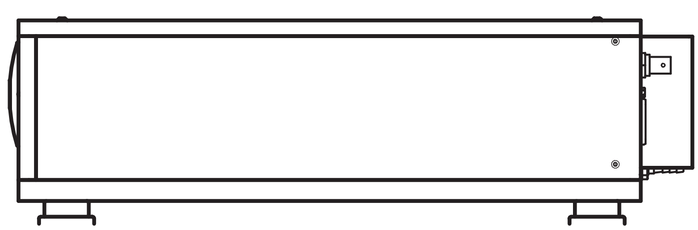
Activating this feedback loop by the switch at the front panel of the master laser controller, the Nd:YAG laser's intensity noise above a few 100 Hz is suppressed by up to 40 dB.

3.5 Mephisto MOPA Laser Head

The laser beam is emitted from the laser aperture at the front side of the laser head. The exact location of the aperture is marked in Figure 3-4 and Figure 3-5. The laser beam is emitted at a right angle with respect to the front side of the laser housing. The aperture can be closed with a mechanical shutter. The handle of the shutter is located at the side of the laser housing. Moving it towards the housing closes the laser aperture. Dimensions for the Mephisto MOPA laser head are illustrated in Figure 3-4 and Figure 3-5.

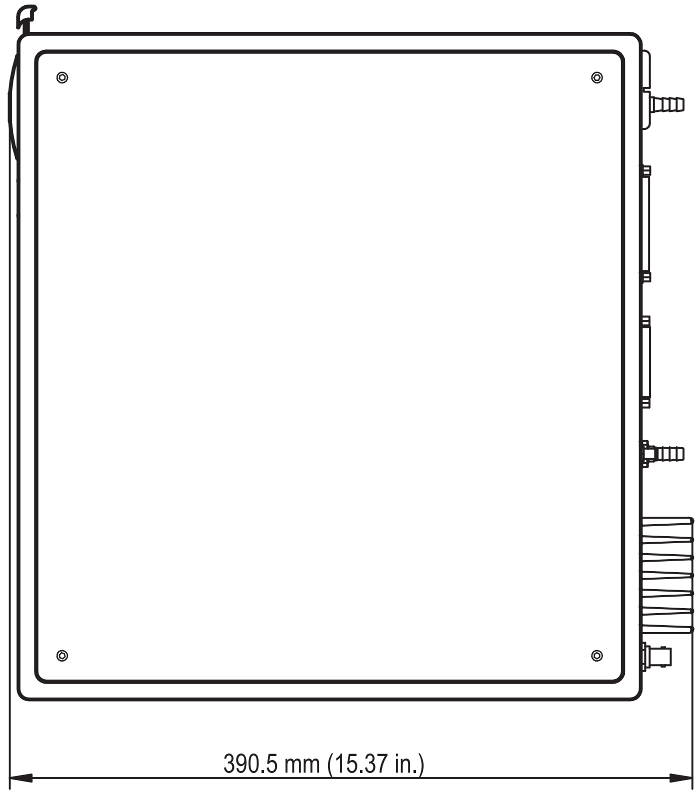


Front View



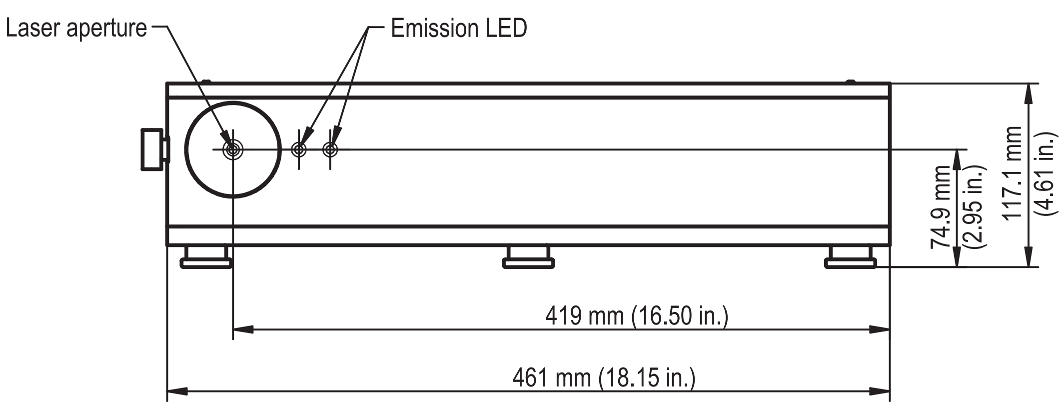
Side View

Figure 3-4. Laser Head Dimensions (8 W, 25 W, or 42 W)



Top Down View

Figure 3-4. Laser Head Dimensions (8 W, 25 W, or 42 W) (Continued)



Front View

Figure 3-5. Laser Head Dimensions (55 W)

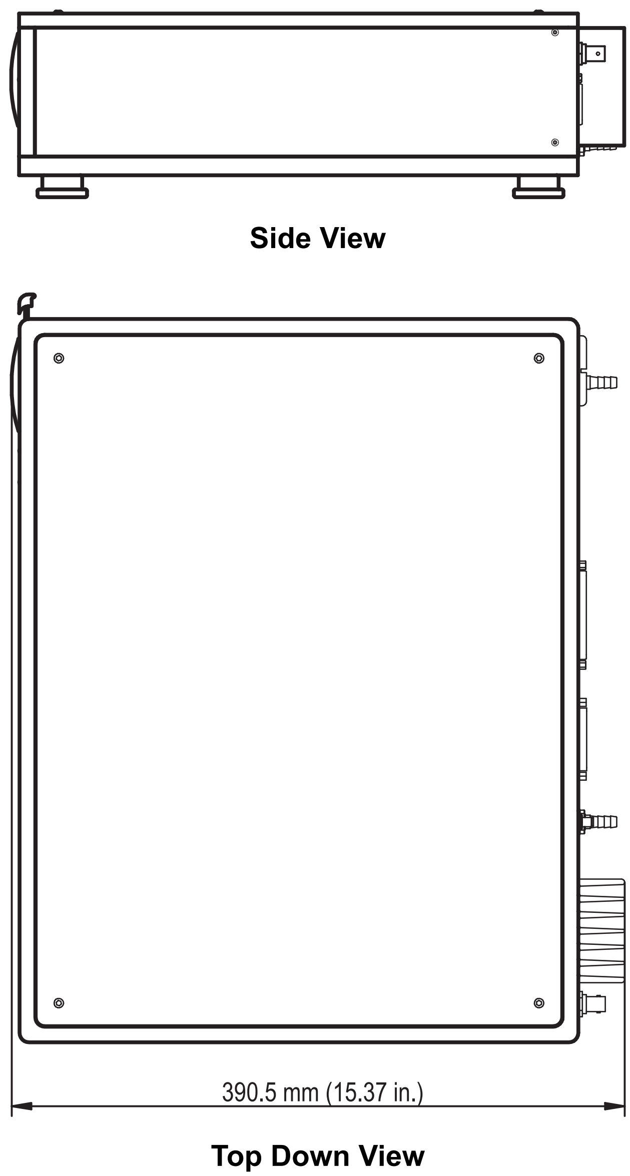


Figure 3-5. Laser Head Dimensions (55 W)

3.6 Master Laser Controller

The four features of the master laser controller include:

- **Laser Diode Driver** - provides a very stable, low noise injection current to the master laser diode up to a value of 3 A. This subsystem also contains the protection circuitry that is essential for reliable operation of the laser system and a temperature controller that regulates the laser diode temperature.

- **Precision Temperature Controller** - stabilizes the Nd:YAG crystal’s temperature. Because of an integrated pre-stabilization stage, typical drifts of this controller are only a few 100 µK / min, corresponding to a variation of the laser frequency of less than 1 MHz / min.
- **Analog modulation inputs (BNC connectors)** - used for laser diode current and laser crystal temperature to externally control output power and laser frequency of the Mephisto laser. (An additional modulation input for fast frequency modulation with a piezo crystal is located at the laser head.)
- **Diagnostics connector (D-Sub connector)** - monitors all vital signals and voltages of the master laser without opening the controller.

Front Panel

Rear Panel

Figure 3-6. Master Laser Controller

3.7 Amplifier Controller

The three features of the amplifier controller include:

- **Laser Diode Driver** - provides a very stable, low noise injection current to the amplifier laser diodes up to a value of 60 A.
- **Precision Temperature Controllers** - stabilize the amplifier laser diode's temperatures.
- **Diagnostics connector (D-Sub connector)** - monitors all vital signals and voltages of the Mephisto MOPA amplifier without opening the amplifier controller.

Static charge is also a well known killer of laser diodes. For this reason, when the laser head is not connected to the controller, a built-in relay shorts the diode pins. After the controller is connected and switched on, the relay opens the short circuit.

Front Panel

Rear Panel

Figure 3-7. Amplifier Controller

The pin configuration of the laser and control connectors are described in “Pin Configurations” (p. 55).

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4 INSTALLATION

4.1 Installing the Mephisto MOPA Laser System

When the product is received, the shipping container and its content must be inspected for any damage incurred during shipping or transit. In case of damage, please notify Coherent and a representative of the carrier immediately!

NOTICE
Keep the original shipping containers! The laser should be shipped only in the packing provided by Coherent with the laser to prevent mechanical shock during transport.

NOTICE
To avoid damage, allow 4 hours for the laser to acclimate before opening the package to avoid condensation!

CAUTION!
The laser head is heavy and weighs 26 or 28 kg (57.32 or 61.73 lbs.), depending on model. It should only be lifted by at least two people to avoid injury.

For accurate installation of the Mephisto MOPA laser system, the following installation sequence is required:

1. Place the laser head on a flat surface, for instance an optical table. For mounting the Mephisto MOPA laser head, use the three clamping forks that are shipped with the laser to mount the pedestals to the optical table. Further clamping or squeezing of the laser head is not recommended, as it may cause temporary misalignments of the optics inside.

2. The heat sink of the laser and the air flow of the controllers should not be blocked or covered. Place them on the table with free surroundings.

NOTICE
To avoid damage, do not place the laser head or the electronics sideways or upside down. Do not place anything on the laser head or on the electronics.

WARNING!
Before connecting the power cable, verify that the line voltage setting on the reverse side of the controllers agrees with local line voltage regulations to avoid damage to the controllers or causing bodily injury.

3. Only use a certified power cable (type H05VV-F; 3G1,0, connector C13 for master laser controller, connector C19 for amplifier controller, no longer than 3.5m). Do not replace the shipped power cable with a cable that is insufficient for this application. Only unplugging the cable completely removes the laser from the line voltage. Therefore, access to the connectors should not be blocked. Only plug the system into a main socket outlet with a protected earth.
4. For activating the laser diode drivers, the two left laser disable pins of the three-pole screw-type connectors at the rear of the controllers must be connected. The laser disable circuit should also include the alarm interlock of the applied water-cooling system. This is to ensure that the amplifier stage of the Mephisto MOPA laser system will shut down in case of water-cooling system failure. With an unconnected laser disable, the unit will be inoperable.

NOTICE
This interface must not be used for any safety functions. Additional controls that disconnect the supply voltage are required.

5. The three units of the Mephisto MOPA laser system, the laser head and the two controllers, are to be connected only by using the suitable cables shipped with the laser system. Make sure that the cables are fully plugged in and secured by the screws. The controller is not to be used with any device other than the laser head it is shipped with. Only turn on the controllers after the cables are plugged in. Do not loop the cables. Turn off the controllers and wait 1 minute for the capacitors to discharge before disconnecting the cables.

WARNING!
To avoid electrical shock, DO NOT touch the cable pins and the connector pins.

NOTICE
Before connecting the two units, make sure the controllers are switched off to avoid damage to the laser.

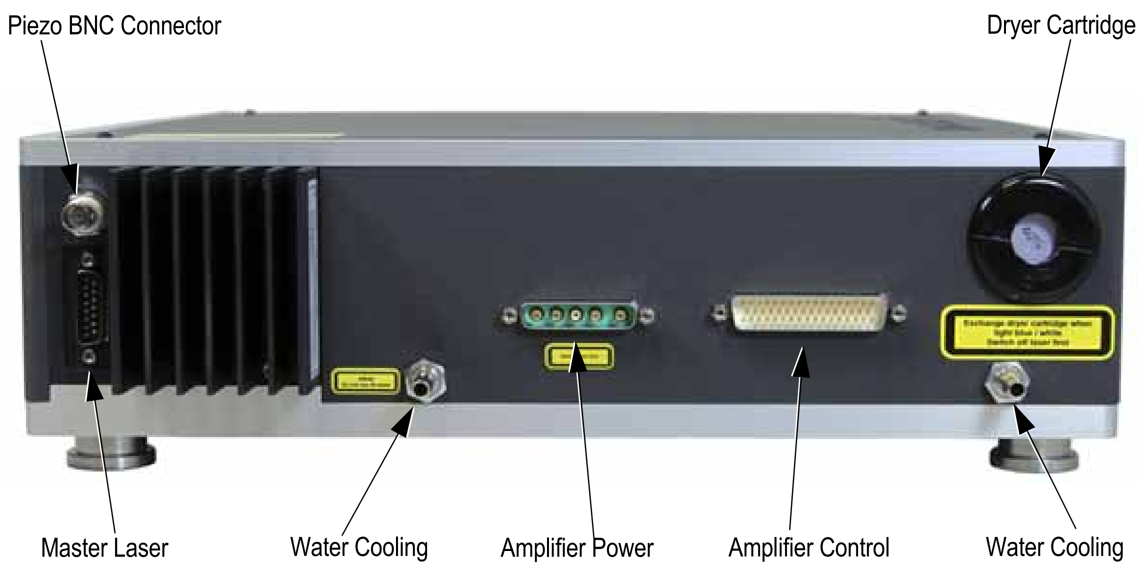

Figure 4-1. Rear View of the Mephisto MOPA Laser Head

At the left of the rear side of the laser head (see Figure 4-1), a D-Sub connector and a BNC connector are located. The D-Sub connector is used to connect the master laser with the controller (the corresponding connector at the controller is labeled laser). The BNC connector can be used to apply a high voltage signal in the range - 65 V to + 65 V to a piezo element on the laser crystal for fast tuning of the laser frequency.

To connect the amplifier with the controller there is a 5 pin high current D-Sub connector (labeled amplifier power at the controller) and a 50 pin D-Sub connector (labeled amplifier Ctrl at the controller). Both are located at the lower center of the rear side of the laser unit. Furthermore, there are two water-cooling plugs and a dryer cartridge is attached for regulating relative humidity inside the laser head.

6. For applying the Mephisto MOPA laser system to a water-cooling system, a cooling power of about 300 W is sufficient for a laser with up to 25 W output power in a common laboratory environment with a room temperature of 20 °C to 25 °C (68 ºF or 77 ºF). If the output power of the Mephisto MOPA laser is up to 42 W, a cooling power of 450 W is sufficient, at 55 W an output power of 600 W is needed. Non-deionized water with a corrosion inhibitor and algae control additives is recommended. Metals inside the laser that are in contact with the water consist of copper, tin, messing, and nickel. The recommended water flow through the laser is > 3 l/min at a pressure of 4 bar. A pressure drop in between the cooler and the laser head should be taken into account if both separated by several meters. The pipe used needs to have an inside diameter of 6 mm and should be connected to the laser head by metal clamps. The pressure should not exceed 6 bar. A certified pressure valve with a maximum of 6 bar must be included in the cooling circuit. The recommended cooling-water temperature for the common laboratory environment, described above, is 25°C (77ºF). It is necessary to check that the water temperature is well above the dew point of the laser's environment. The maximum environmental temperature is 30ºC (86ºF).

The permissible range of the cooling water temperature is 20ºC (68ºF) to 30ºC (86ºF). However, depending on the relative humidity and the temperature of the ambient air, it is crucial not to operate the cooling water temperature below a certain limit to avoid condensation. The specific lower limit depending on local ambient air conditions can be obtained from the table below. Running the cooling water temperature below the specific lower limit will void the laser system warranty! Once condensation has occurred, the optic inside the laser head cannot be cleaned by the operator and will have to be shipped back to Coherent for repair.

The permissible range of the laser crystal temperature of the master laser is 20ºC (68ºF) to above 40ºC (104ºF). However, depending on the relative humidity and the temperature of the ambient air, condensation must also be avoided for the master laser crystal. The specific lower limit for the laser crystal temperature depending on the ambient air conditions can also be obtained from Table 4-1. Operating the laser crystal temperature below the specific lower limit will void the laser system warranty!

Table 4-1. Permissible Minimum Cooling Water Temperature ºC (ºF) for Different Relative Humidity Conditions

Ambient Air Temperature ºC (ºF)	≤ 40%RH	45%RH	50%RH	55%RH	60%RH	65%RH	70%RH	75%RH	80%RH	≥ 80%RH
< 15 (59)										
15 (59)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	
16 (60.8)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	
17 (62.6)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	
18 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	
19 (66.2)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.5 (65.3)	
20 (68)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.5 (65.3)	19.5 (67.1)	
21 (69.8)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.4 (65.12)	19.4 (66.92)	20.4 (68.72)	
22 (71.6)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.2 (64.76)	19.3 (66.74)	20.4 (68.72)	21.4 (70.52)	
23 (73.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	19.0 (66.2)	20.3 (68.54)	21.4 (70.52)	22.4 (72.32)	
24 (75.2)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.7 (65.66)	20.0 (68)	21.2 (70.16)	22.2 (71.96)	23.3 (73.94)	
25 (77)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	18.4 (65.12)	19.7 (67.46)	21.0 (69.8)	22.1 (71.78)	23.2 (73.76)	24.4 (75.92)	
26 (78.8)	18.0 (64.4)	18.0 (64.4)	18.0 (64.4)	19.3 (66.74)	20.7 (69.26)	21.9 (71.42)	23.1 (73.58)	24.3 (75.74)	26.3 (79.34)	
28 (82.4)	18.0 (64.4)	18.0 (64.4)	19.6 (67.28)	21.1 (69.98)	22.4 (72.32)	23.9 (75.02)	25.0 (77)	26.2 (79.16)	27.2 (80.96)	
30 (86)	18.0 (64.4)	19.8 (67.64)	21.4 (70.52)	23.0 (73.4)	24.4 (75.92)	25.7 (78.26)	26.9 (80.42)	28.1 (82.58)	29.1 (84.38)	
> 30 (86)	Don't operate and cool the laser system at these ambient air conditions									

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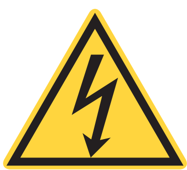
5 OPERATION

NOTICE
Read this manual carefully before starting up the laser! Operating errors can lead to destruction of the Mephisto MOPA laser head!

NOTICE
Before connecting or disconnecting any cables, switch off the control electronics and/or the power supply to avoid damaging the laser!



WARNING!
To avoid electrical shock, do not open the controllers!

WARNING!
Always wear laser safety glasses to protect the eyes! Those using the laser and everyone else in the room must wear safety glasses as required by the wavelength being generated to protect against accidental exposure to the output beam or its reflection.

5.1 Turning the Laser On

For proper operation of the Mephisto MOPA laser system, the following turn-on sequence is recommended:

1. Make sure the water cooling circuit is activated and wait for the cooling system to stabilize.
2. Use the main keyswitch to turn the master laser controller and amplifier controller on. The OFF button of each controller will glow and the fan will be operating.
3. Check the Set Temperature of the master laser diode at the diagnostics connector. Compare the values with those given in the data sheet of the laser.
4. Check the Set Temperature of the Nd:YAG laser crystal. The correct operation temperature depends on the experimental requirements and may be around 25ºC (77ºF).
5. Allow about 60 seconds for the temperature controllers to stabilize, particularly before starting to adjust the Nd:YAG laser crystal temperature to higher values.
6. Check the actual temperature of the amplifier laser diodes by pushing the Actual button on the front of the amplifier controller. Compare the values with those given in the data sheet of the laser.
7. Make sure the “error” LED at the front of the master laser controller as well as all LEDs at the front of the amplifier controller are not glowing.
8. Check the set current at the diagnostics connector of the master laser controller.
9. Activate the master laser diode driver by pressing the ON button. The laser emission LEDs at the front of the laser head will glow. Make sure that the light from the LEDs is still visible through the laser safety glasses.
10. Place a suitable power detection device or a beam dump in front of the laser aperture and open the shutter on the side of the laser head.
11. Activate the amplifier diode driver by pressing the green ON button. The fan will increase its air stream and the button will glow green.
12. Change the Injection Current of the power amplifier until the desired value is displayed at the amplifier display (see data sheet).

NOTICE

To avoid damage to the laser do not turn on the amplifier without having turned on the master laser first!

NOTICE

Do not operate the amplifier without water cooling to avoid damage to the laser!

To reduce the MOPA output power, the amplifier current can be reduced, however this changes the beam divergence.

NOTICE
Do not reduce the master laser current to avoid damage to the laser!

NOTICE
To avoid damage to the laser and not to disturb the laser performance, avoid back-reflections from the optical set-up into the MOPA laser. If such reflections should occur, use suitable optics like a Faraday Isolator with sufficiently high optical isolation.

To protect the laser diodes, a current limiting circuitry is included. The Injection Current is restricted to the range from zero to the internally set current limit.

Both controller units feature a soft-start that smoothly increases the laser diode current when the LASER ON / OFF buttons are pressed.

5.2 Turning the Laser Off

Go through the following steps to switch off the Mephisto MOPA laser system:

1. Deactivate the amplifier by pressing the red OFF button. The button will glow red.
2. Turn off the amplifier controller with the main keyswitch.
3. Deactivate the master laser by pressing the red OFF button. The button will glow red.
4. Switch off the master laser controller with the main keyswitch.

NOTICE
Do not turn off the master laser before the amplifier has been turned off to avoid damage to the laser!

NOTICE
Lasers may be damaged by improper setting of the current controls or by improper use of the modulation inputs.

5.3 Recommended Operation

The optical set-up inside the laser head is carefully aligned for best performance. Therefore, the laser head should be handled with caution. Any mechanical shock is hazardous! Misalignment will cause the optical power to decrease and should not be fixed by the operator! Check the output power using a calibrated power meter.

In case of low optical output power or poor beam quality, do not attempt to fix the problem. Please contact Coherent immediately!

WARNING!
Hazardous amount of invisible laser radiation may be diffracted in any direction. Always use protective eyewear when operating the Mephisto MOPA!

Check the functionality of the laser emission LED at regular intervals.

The laser can be operated continuously. It can be operated with the shutter fully opened and with the shutter fully closed.

Whenever the laser is not used, close the mechanical shutter to avoid accidental exposure to laser radiation.

NOTICE
Be aware that the laser might disturb equipment that is sensitive to magnetic fields.

5.4 Frequency Tuning Capabilities

The frequency of the Mephisto MOPA laser can be tuned by changing the temperature of the monolithic laser crystal. This can either be done directly at the front panel of the master laser controller using the appropriate trimmer or by applying a voltage to the laser crystal temperature modulation input at rear of the controller.

By applying an analog voltage signal in the range - 10 V to + 10 V to the modulation input labeled “Temperature Laser Crystal”, the temperature of the laser crystal can be changed by + 1 K/V, corresponding to a frequency change of about - 3 GHz/V at 1064 nm. Due to the large time constants of the thermal tuning, the response bandwidth is limited to fractions of a Hertz. The input impedance is 1 kΩ. Do not exceed +/- 10 V and avoid condensation.

The typical tuning characteristic of a standard Mephisto laser operating at 1064 nm is shown in Figure 5-1. The filled dots represent operation on a single longitudinal frequency, while the open dots indicate the mode-hops, where the laser frequency changes from one longitudinal mode to the next. The exact position of the mode-hops can be shifted a little from one laser to the next.

As can be seen, the laser frequency can be tuned continuously by 6 - 8 GHz between the mode-hops, covering an overall frequency tuning range of more than 30 GHz.

Fast frequency tuning of a Mephisto laser can be achieved by applying a high voltage signal in the range of - 65 V to + 65 V to a piezo crystal element on the laser crystal, using the BNC connector at the rear side of the laser head. Higher voltages will misalign the laser cavity or even destroy the piezo crystal. Especially RF signals will heat up the piezo and destroy it.

Depending on the actual laser crystal and the modulation frequency, the piezo tuning coefficient is about 1 MHz/V with a response bandwidth of about 100 kHz. The capacity is 1 - 2 nF. The combination of the slow temperature tuning with a large range and the fast tuning with a high bandwidth is ideally suited for stabilizing the laser frequency to optical reference cavities or molecular absorption lines.

5.5 Output Power Modulation

By applying an analog voltage signal in the range - 10 V to + 10 V to the modulation input labeled “Current Laser Diode” using the BNC connector at the rear panel of the master laser controller, the injection current of the master laser can be modulated by 0.1 A/V. The response bandwidth of this modulation input is limited to about 5 kHz, the input impedance is

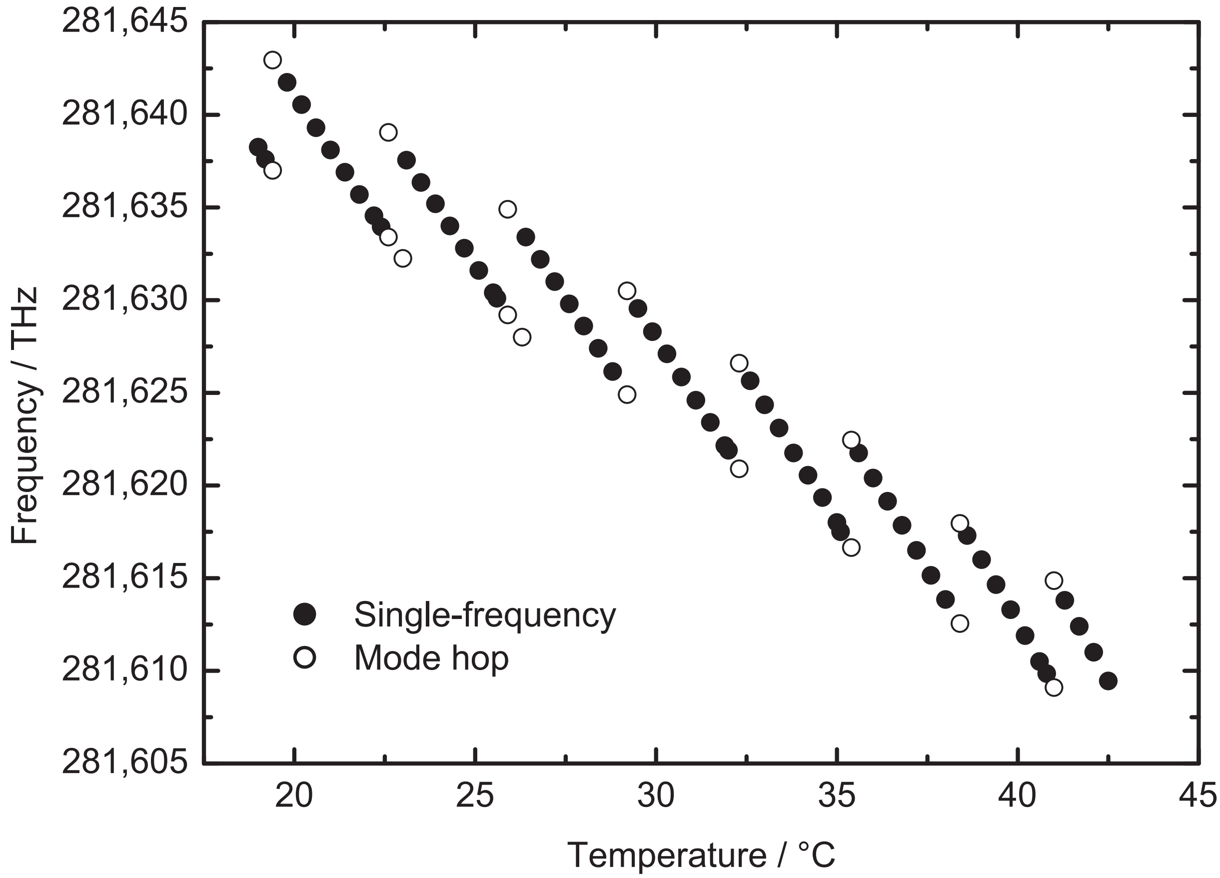


Figure 5-1. Typical Frequency Tuning Over Crystal Temperature

100 kΩ. Do not exceed +/- 10 V. Because of a nonlinear amplification factor it is not recommended to use this option for modulating the overall output power. Also, the master laser should not be modulated significantly at DC.

A BNC connector at the backside of the amplifier controller allows a modulation of the amplifier current by 10 A/V. Do not exceed +/-10 V. The modulation bandwidth is up to 10 Hz (5 kΩ input resistance). Only very small modulations should be applied here. A big DC change in the amplifier current results in changing thermal properties inside the laser. This leads to changes in the beam size.

6 TROUBLESHOOTING

6.1 Introduction

If any failure occurs, do not open the laser head or the controllers! They do not contain any user serviceable parts.

To avoid damage to the laser or bodily injury, do not use any replacement parts that are not authorized by Coherent.

Be aware that laser diodes degrade and have a limited lifetime. In case of a reduced output power, contact Coherent to determine if the diodes have degraded. If that is the case, a diode exchange at Coherent might be necessary.

The controllers for the Mephisto MOPA laser system include a sophisticated safety circuitry to protect the laser diodes against current spikes or overheating and switch the diode drivers into inactive mode in case of problems. This is indicated by the LEDs at the front panel of the master laser and amplifier controller. Check for the following possible causes:

6.1.1 Laser Disable (Master Laser Controller)

Diagnosis:
The two pins of the laser disable connector at the rear panel of the controller are not connected.

Reaction:
Short the two pins of the laser disable connector.

6.1.2 Laser Disable (Amplifier Controller)

Diagnosis:
The two pins of the laser disable connector at the rear panel of the controller are not connected.

Reaction:
Check the water cooling system for malfunction and the interlock switches of the water chiller. Short the two pins of the laser disable connector.

6.1.3 Temperature Guard

Diagnosis:
Indicated by the "Error" LED (master laser), "Guard" LED's (amplifier). To prevent the laser diodes from overheating, the controller contains a Temperature Guard that monitors deviations between the set temperature and the actual laser diode temperature and switches the driver into inactive mode in case a deviation persists.

Reaction:
If the set temperature is far above the room temperature, the temperature control might not reach the set temperature quick enough before the temperature guard is activated. Turn the controller on and off. Starting from a preheated value, the set temperature will now be reached more quickly.

If the set temperature is not particularly high and is not reached, it might be due to overheating of the laser head. Check the cooling water temperature and the flow rate. Also check that the heat sink is not blocked. If the set temperature was set far too low and cannot be reached, try increasing it a little.

If the problem persists, contact Coherent.

6.1.4 Clamp Current

Diagnosis:
Indicated by the "Error" LED (master laser), "Clamp" LED (amplifier). The injection current of the laser diodes is driven above its internal limit.

Reaction:
Reduce the voltage at the "Current Laser Diode" modulation input at the rear panel of the master laser controller or at the "Amplifier Modulation In" input of the amplifier laser controller.

6.1.5 Heat Sink Error

Diagnosis:
Indicated by the "Heat Sink Error" LED. The base plate inside the laser head is getting too warm or too cold.

Reaction:
Check the water cooler. The flow rate might be too low or the water temperature might be too high or too low.

After fixing an error, cycle the keyswitch once to see if the error is no longer present. If the problem cannot be fixed, contact Coherent. Service inside the laser head or the controller should only be performed at Coherent.

6.1.6 Amplitude Noise

If there are big high frequency oscillations in the laser amplitude noise, turn off and on the Noise Eater to see if the oscillation disappears.

If the problem cannot be fixed, contact Coherent. Service inside the laser head or the controller should only be performed at Coherent.

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7 MAINTENANCE

7.1 Drying Cartridge

The drying cartridge provides a constant low relative humidity of 40% inside the laser and is located at the rear side of the laser head. It must be checked in three month intervals to ensure accurate operation. If the color of the optical indicator has turned from blue to completely white, it must be changed.

The installed drying cartridge has a metric thread M 36 x 1.5, a mounting dimension of 45 mm and is sufficient for a whitespace of 100 dm³. To be provided with suitable drying cartridges, contact Coherent. Make sure that the master laser and the amplifier are turned off when exchanging the cartridge!

WARNING!
To avoid hazardous invisible laser radiation, the Mephisto MOPA laser system has to be turned off to replace the drying cartridge.

Aside from the drying cartridge, the Mephisto MOPA laser head and the controllers do not contain any parts requiring maintenance. Do not open the laser head or the controller.

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8 PIN CONFIGURATIONS

8.1 Master Laser Connector

The laser head and the master laser controller are to be connected only with suitable cables, shipped with the laser system. Figure 8-1 illustrates the 37 pin D-Sub connector at the rear panel of the master laser controller. The description of the individual pins of the laser connector is given in Table 8-1 (p. 55).

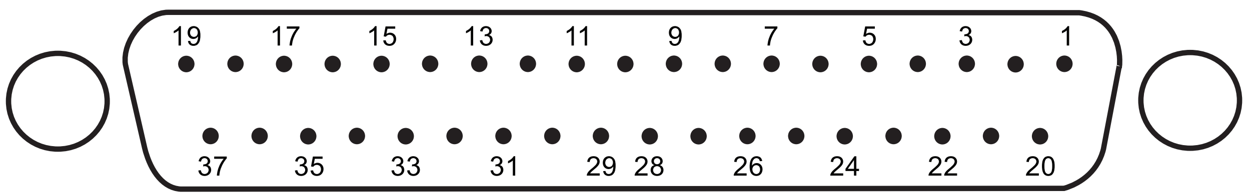

Figure 8-1. Master Laser Connector (37 pin D-Sub)

Table 8-1. Pin Description - Master Laser Connector

Pin	Description	Pin	Description
1	Laser diode cathode	20	Laser diode anode
2	Laser diode, TEC anode	21	Supply voltage + 12 V
3	Laser diode, TEC cathode	22	Relay, negative supply voltage
4	n/c	23	Connected to 37
5	n/c	24	Relay, positive supply voltage
6	Laser crystal, TEC anode	25	Supply voltage - 12 V
7	Laser crystal, TEC cathode	26	n/c
8	n/c	27	Noise Eater, switch
9	n/c	28	n/c
10	n/c	29	n/c
11	n/c	30	n/c

Table 8-1. Pin Description - Master Laser Connector (Continued)

Pin	Description	Pin	Description
12	NTC reference voltage 6.85 V	31	n/c
13	Laser crystal, NTC ground	32	n/c
14	n/c	33	n/c
15	NTC reference voltage 6.85 V	34	n/c
16	Laser diode, NTC ground	35	- 5 V supply, Noise Eater
17	n/c	36	+ 5 V supply, Noise Eater
18	+5V	37	Connected to 23
19	GND		

8.2 Master Laser Diagnostics Connector

All vital information about the status of the Mephisto MOPA laser system can be monitored without opening the controller, using the diagnostics connectors. Figure 8-2 illustrates the 25 pin D-Sub connector at the rear panel of the master laser controller:

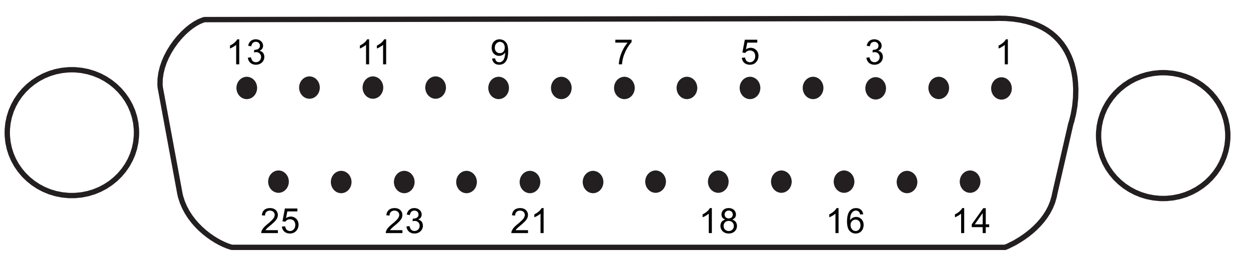

Figure 8-2. Master Diagnostics Connector (25 pin D-Sub)

A description of the individual pins of the diagnostics connector for the master laser controller is given in Table 8-2. These signals should be monitored with > 10 kΩ load:

Table 8-2. Pin Description - Master Diagnostics Connector

Pin	Description
1	Laser diode current limit (1 A / V)
2	Laser diode actual current (1 A / V)
3	Laser diode set temperature (0.1 °C / mV)
4	Laser diode actual temperature (0.1 °C / mV)
5	Laser diode temperature control error signal (settles to 0 after warm-up)
6	Laser crystal set temperature (0.1 °C / mV)
7	Laser crystal actual temperature (0.1 °C / mV)
8	Laser crystal temperature control error signal (settles to 0 after warm-up)
9	Error indicator (positive voltage up to 12 V: no error, negative voltage up to -12 V: error)
10	Do not use
11	Do not use
12	Do not use
13	Do not use
14-25	GND (measure all other signals against GND)

8.3 Amplifier Connector

The laser head and the amplifier controller are to be connected only with the suitable cables shipped with the laser system. Figure 8-3 illustrates the 5 pin D-Sub connector at the rear panel of the amplifier controller:

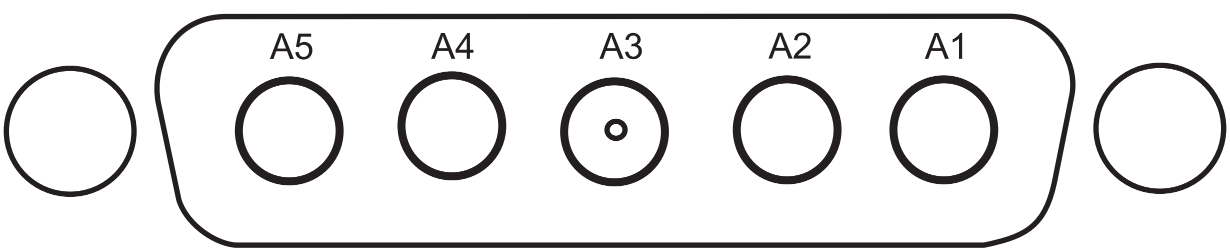

Figure 8-3. Amplifier Current Connector (5 pin D-Sub)

The description of the individual pins of the amplifier current connector is given in Table 8-3:

Table 8-3. Pin Description - Amplifier Current Connector

Pin	Description
A1	Laser diode cathode
A2	Laser diode cathode
A3	Safety relay control
A4	Laser diode anode
A5	Laser diode anode

8.4 Amplifier Control Connector

In addition, there is a 50 pin D-Sub connector (Figure 8-4). A description of the individual pins of the control connector for the amplifier controller is given in Table 8-4.

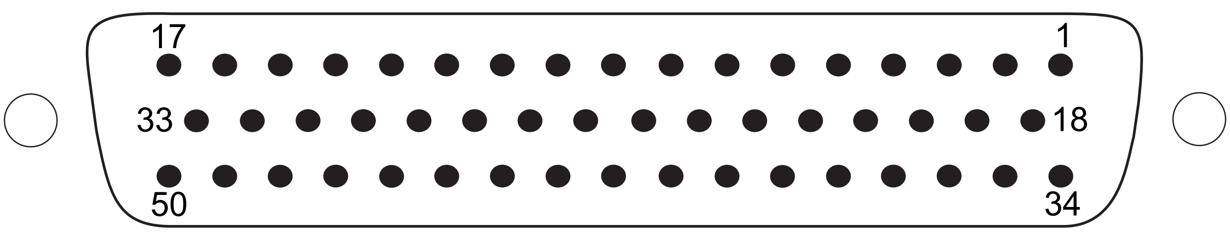

Figure 8-4. Amplifier Control Connector (50 Pin D-Sub)

Table 8-4. Pin Description - Amplifier Control Connector

Pin	Description
1	+5V_safe1
2	GND_safe1
3	GND
4	GND
5	Spare1
6	Spare2
7	n/c
8	GND
9	TEC-1-
10	TEC-1+
11	TEC-2-
12	TEC-2+
13	TEC-3-
14	TEC-3+
15	TEC-4-
16	TEC-4+
17	+5V_safe2
18	n/c
19	n/c
20	GND
21	n/c
22	n/c
23	n/c
24	n/c
25	n/c

Pin	Description
26	TEC-1-
27	TEC-1+
28	TEC-2-
29	TEC-2+
30	TEC-3-
31	TEC-3+
32	TEC-4-
33	TEC-4+
34	NTC-4
35	NTC-3
36	NTC-2
37	NTC-1
38	NTC-HS
39	NTC-RET
40	MON-LD-1
41	MON-LD-2
42	MON-LD-3
43	MON-LD-4
44	MON-NE
45	NE ON/OFF
46	Master Error
47	GND_Safe2
48	- 15 V
49	GND
50	+ 15 V

8.5 Amplifier Diagnostics Connector

Figure 8-5 illustrates the 25 pin D-Sub connector at the rear panel of the amplifier controller:

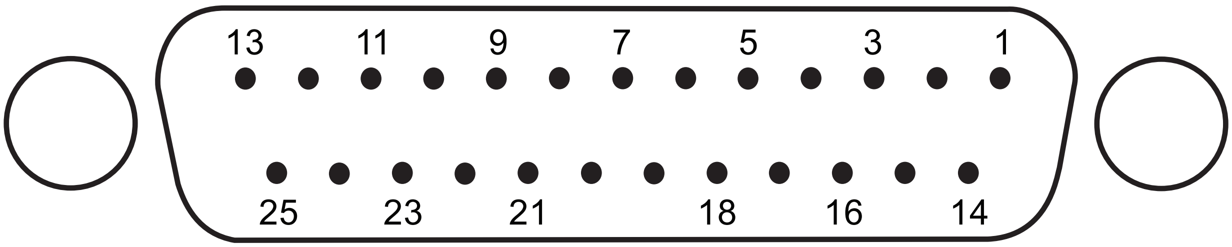

Figure 8-5. Amplifier Diagnostics Connector (25 pin D-Sub)

A description of the individual pins of the diagnostic connector for the amplifier controller is given in Table 8-5. These signals should be monitored with > 10 kΩ load:

Table 8-5. Pin Description - Amplifier Diagnostics Connector

Pin	Description
1	Actual Current (6 A / V)
2	Amplifier on monitor (on: 5 V, off: 0 V)
3	Do not use
4	Diode 1 temperature control error signal (settles to 0 after warm-up)
5	Diode 2 temperature control error signal (only in 2, 3 and 4 diode systems, settles to 0 after warm-up)
6	Diode 3 temperature control error signal (only in 3 and 4 diode systems, settles to 0 after warm-up)
7	Diode 4 temperature control error signal (only in 4 diode systems, settles to 0 after warm-up)
8	Actual temperature heat sink (0.1 °C / mV)
9	Do not use
10	Do not use
11	Do not use
12	Do not use
13	Laser disable monitor (closed: 0 V, open: 5 V)
14	Do not use
15-25	GND (measure all other signals against GND)

I ACCESSORIES

I.1 Power Meter Accessories

Coherent offers a variety of instruments for laser test and measurement. For additional detailed information, including product selection guides, visit our web site at www.coherent.com.

NOTICE

Ensure that maximum power density of the power sensor is not exceeded by the laser beam irradiance.

I.1.1 Coherent's Recommendation

For the most common diagnostics – measuring the output power of the Mephisto – Coherent recommends the Fieldmax™II-TO with a PM-30 sensor for Mephisto MOPA lasers < 30 W and a PM-150 sensor for Mephisto MOPA lasers > 30 W (see Figure I-1). The FieldmaxII-TO is a full-featured power meter that supports interchangeable power sensors and offers capabilities like onboard statistical analysis and computer interfacing via USB. It also comes with installable applications software and LabVIEW drivers. This full feature power meter set allows for advanced power measurement including the following features:

- Large, bright, backlit LCD display
- Digital accuracy with analog-like movement for laser tuning
- Works with thermopile and optical sensors only
- Measures pulse energy up to 300 pps (captures every pulse)
- USB computer interface with complete host control capability (Windows Vista and Windows 7 [32 and 64 bit] compatible)
- Analog output with user selectable full-scale voltage (1,2, and 5 V)
- Rechargeable battery included
- RoHS compliant

FieldmaxII-TO Laser Power Meter
Part Number 1098579

PM30 Power Sensor (Mephisto MOPA < 30 W)
Part Number 1098314

PM150 Power Sensor (Mephisto MOPA > 30 W)
Part Number 1098407

Figure I-1. Coherent Power/Energy Meter & Sensor (Recommended)

I.2 Drying Cartridges

Additional drying cartridges for the Mephisto MOPA can be purchased from Coherent (Part Number 1278526).

I.3 Water Cooler

A water cooler suitable for the Mephisto MOPA laser with all the necessary accessories, such as connectors can be purchased from Coherent.

Laser System	Part Number
MOPA 8 W and 25 W	1279585
MOPA 42 W and 55 W	1279586

II WARRANTY

Coherent, Inc. warrants Diode-Pumped Solid State laser systems to the original purchaser (the Buyer) only, that the laser system, that is the subject of this sale, (a) conforms to Coherent's published specifications and (b) is free from defects in materials and workmanship.

Laser systems are warranted to conform to Coherent's published specifications and to be free from defects in materials and workmanship for thirteen (13) months.

II.1 Responsibilities of the Buyer

The buyer is responsible for providing the appropriate utilities and an operating environment as outlined in the product literature. Damage to the laser system caused by failure of buyer's utilities or failure to maintain an appropriate operating environment, is solely the responsibility of the buyer and is specifically excluded from any warranty, warranty extension, or service agreement.

The Buyer is responsible for prompt notification to Coherent of any claims made under warranty. In no event will Coherent be responsible for warranty claims made later than seven (7) days after the expiration of warranty.

II.2 Limitations of Warranty

The foregoing warranty shall not apply to defects resulting from:

- Components and accessories manufactured by companies, other than Coherent, which have separate warranties
- Improper or inadequate maintenance by the buyer
- Buyer-supplied interfacing
- Operation outside the environmental specifications of the product
- Unauthorized modification or misuse
- Improper site preparation and maintenance

- Opening the housing

Coherent assumes no responsibility for customer-supplied material. The obligations of Coherent are limited to repairing or replacing, without charge, equipment which proves to be defective during the warranty period. Replacement sub-assemblies may contain reconditioned parts. Repaired or replaced parts are warranted for the duration of the original warranty period only. The warranty on parts purchased after expiration of system warranty is ninety (90) days. Our warranty does not cover damage due to misuse, negligence or accidents, or damage due to installations, repairs or adjustments not specifically authorized by Coherent.

Warranty applies only to the original purchaser at the initial installation point in the country of purchase, unless otherwise specified in the sales contract. Warranty is transferable to another location or to another customer only by special agreement which will include additional inspection or installation at the new site. Coherent disclaims any responsibility to provide product warranty, technical or service support to a customer that acquires products from someone other than Coherent or an authorized representative.

THIS WARRANTY IS EXCLUSIVE IN LIEU OF ALL OTHER WARRANTIES, WHETHER WRITTEN, ORAL OR IMPLIED, AND DOES NOT COVER INCIDENTAL OR CONSEQUENTIAL LOSS. COHERENT SPECIFICALLY DISCLAIMS THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

GLOSSARY

%	Percent
°C	Degrees Celsius
°F	Degrees Fahrenheit
µ	Micron(s) = (10^{-6})
µJ	MicroJoule(s) = (10^{-6} Joules)
µK	Microkelvin(s) = (10^{-6} Kelvin)
µm	Micrometer(s) = (10^{-6} meters)
Ω	Ohm (SI derived unit of electrical resistance)
A	Ampere(s)
AC	Alternating Current
a.u.	Arbitrary units
BNC	Bayonet Neill Concelman connector
CDRH	Center for Devices and Radiological Health (U.S. Government)
cm	Centimeter(s) = (10^{-2} meters)
dB	Decibel
DC	Direct Current
dm	Decimeter
EMC	Electromagnetic Compatibility
F	Farad(s)
FPI	Fabry-Pérot Interferometer
FSR	Free spectral range
GHz	GigaHertz = (10^{9} Hertz)
GND	Ground
hPa	Hectopascal
HS	Heatsink
Hz	Hertz or cycles per second (frequency) (= 1/pulse period)
in.	Inch(es)
IP	Ingress protection rating
ISO	International Organization for Standardization
K	Kelvin(s)
kHz	Kilohertz = (10^{3} hertz) (1000 hertz)
kg	Kilogram(s) = (10^{3} grams)
km	Kilometer(s) = (10^{3} meters)
kΩ	Kilohm
LD	Laser Diode
LED	Light Emitting Diode
m	Meter(s) (length)

mA	Milliamp(s) = (10^{-3} Amperes)
m°C	Milidegree(s) Celsius
MHz	Mega Hertz = (10^{6} Hertz)
mm	Millimeter(s)
MON	Monitor
MOPA	Master Oscillator Power Amplifier
MΩ	Megohm(s) = 10^{6} ohms) (resistance)
mrad	Milliradian(s) = (10^{-3} radians) (angle)
ms	Millisecond(s) = (10^{-3} seconds)
mV	Millivolt(s)
mW	Milliwatt(s) (10^{-6} Watts) (power)
n/c	Not connected
NE	Noise Eater
nF	Nanofarad(s) = (10^{-9} Farad)
nm	Nanometer(s) = (10^{-9} m) (wavelength)
NTC	Negative Temperature Coefficient Thermistor
pps	Captures per cycle
RET	Return
RF	Radio Frequency
RIN	Relative intensity noise
RoHS	Restriction of Hazardous Substances Directive
TEC	Thermo Electric Cooler
TEM	Transverse ElectroMagnetic (cross-sectional laser beam mode)
THz	Terahertz = (10^{12} hertz)
V	Volt(s)
W	Watt(s)
WEEE	Waste Electrical and Electronic Equipment

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Mephisto™ MOPA Laser Operator’s Manual

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